Toni Mateos

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RESUME

Co-creator of Dolby Atmos, from the very initial research and prototypes to the final product, for which we were awarded an Oscar for Scientific and Technical Achievements by the Academy of Motion Pictures.

With a background as a PhD and Postdoc in Theoretical Physics (String Theory & Quantum Gravity), I transitioned from academia to deep tech. My biggest pleasure became turning "hardcore" science into truly useful products that reach as many people as possible.

I joined Dolby following the acquisition of ImmSound, where I was a co-founder and Head of Research. At Dolby, we ensured that Atmos reached over 1 billion people across more than 90 countries, doing the R&D required to expand from the cinema industry into the home, video game, and music industries.

I fell in love with Deep Learning following the seminal 2012 papers, and pushed to build Dolby's internal capabilities. I eventually contributed to the introduction of Deep Learning into Dolby's audio processing pipeline, evolving our traditional digital signal processing foundation.

In recent years, I have devoted my time to research and development of Blockchain core scaling technology, co-founding Freeverse.io, which raised $10M+ in its Series A, and co-founding the LAOS blockchain, where I serve as Head of Research.

I have always been hands-on and interested in building technology with the potential to go beyond being just a good paper. I am most confident leading when also contributing to the core tech, the algorithms, the basic research or the code.

Author of 30+ patents, I love everything around building hard tech, including the human side. I've had the honor and pleasure of building teams from scratch, attracting young and senior international talent, and helping extremely different personalities grow and converge around the joy of building something together.

FURTHER SOURCES

- LinkedIn profile: https://www.linkedin.com/in/toni-mateos-6855767
- Github: https://github.com/tonimateos
- Personal Page: https://tonimateos.github.io/
- Google Scholar profile: https://scholar.google.es/citations?user=mm-etVoAAAAJ
- Patents: https://patents.justia.com/inventor/antonio-mateos-sole

PROFESSIONAL CAREER

- **Senior Staff AI Researcher** at Dolby Laboratories, 2026-now.
- **Head of Research** and Co-founder at LAOS Blockchain, 2024-2026.
- **CTO and Co-founder** at Freeverse.io, Blockchain Scaling Tech, 2019-2025.
- **Director of Research**, Advanced Technology Group, at Dolby Laboratories, 2016-2019.
- **Senior Staff Engineer**, Advanced Technology Group, at Dolby Laboratories, 2012-2016.
- **Co-founder and Chief Scientific Officer** at ImmSound, 2009-2012.
- **Scientific Director** of Audio Group at Barcelona Media Innovation Center, 2008-2012.

ACADEMIC CAREER

- **Postgraduate Degree** in Blockchain Technologies, Polytechnic University of Catalonia (UPC), 2018.
- **Associate Lecturer** at Dept. Technology, Pompeu Fabra University (UPF), 2006-2012.
- **Postdoc** in Acoustic Field Simulation and Audio Spatialization at Barcelona Media, 2006-2007.
- **Postdoc** in String Theory, and Quantum Field Theory / Supergravity duality at Imperial College London, 2004-2006.
- **Doctor in Theoretical Physics**, University of Barcelona, 2004. Ph.D. Thesis on *D-branes, noncommutative theories and gauge/string duality*.
- **CERN** (European Laboratory for Particle Physics), Geneva, Switzerland, Internship, 1999.
- **Degree in Physics** at the University of Barcelona, Spain (first class Honours), 1995-1999.

AI, DEEP LEARNING, PROGRAMMING

- Solid theoretical foundation in Deep Learning (Information Theory, Statistics, Probability) and Deep Learning architectures, including: Convolutional Neural Networks, Recurrent Neural Networks, LSTMs, Bidirectional RNNs, GRUs, Generative Adversarial Networks, Autoencoders, and Transformers.
- Related languages, libraries and ecosystems: Python, PyTorch, Hugging Face, Weights & Biases, GitHub, Docker, CLI, Linux.
- Fascinated user of "Vibe Coding".
- After 25+ years of hands-on experience in programming, from research prototypes to production, I have various degrees of proficiency in other languages and frameworks: Node.js, React, C/C++, Golang, MATLAB, Mathematica, and Solidity.

HUMAN LANGUAGES

- Fully proficient in **English**, **Spanish** and **Catalan**.
- Fluent in **Modern Greek**. Fair understanding of **Italian** and **French**.

FELLOWSHIPS AND AWARDS

- Oscar for Scientific and Technical Achievements as part of the Engineering Team that created Dolby Atmos, 2024.
- Best peer-reviewed paper at AES 136th Convention, 2014.
- Mathematical Physics - Torres y Quevedo Fellowship, Spain, 2006-2008.
- Particle Physics and Astronomy Research Council Fellowship, UK, 2004-2006.
- PhD Fellowship from the *Generalitat de Catalunya*, Spain, 1999-2003.
- Special Mention for undergraduate studies from the U. of Barcelona.
- Ranked 1st in the 1999 Physics graduating class at the University of Barcelona.

SUPERVISION OF ACADEMIC RESEARCH

PhD Thesis Director:

- *3D audio technologies: applications to sound capture, post-production and listener perception*, Giulio Cengarle, University Pompeu Fabra, 2012.

PhD Thesis Tribunal Member:

- *Wavelet-based spatial audio framework; from Ambisonics to wavelets: a novel approach to spatial audio*, Dept. Information and Communication Technologies, UPF, by Davide Scaini, 2019.
- *Parametric array designs and exploitation methods for directivity control of audible sounds*, Department of Engineering, La Salle Ramon Llull University, by Umut Sayin, 2012.

Master Thesis Supervisor:

- *3D Audio effects for multi-channel reproduction*, Master of Sound and Music Computing, UPF, by Antonio Escamilla, 2009-2010.

Degree Final Projects Supervisor:

- *Musical applications of object-oriented 3D audio*, UPF, by Aleix Fabra Roca, 2008.
- *Tools for 3D audio post-production*, UPF, by Ferran Orriols Lpez, 2008.
- *Livecoreo, a graphical tool for 3D sound post-production*, UPF, by Angelo Scorza, 2011.

TEACHING EXPERIENCE

- *Acoustics Engineering*, Faculty of Audiovisual Engineering, University Pompeu Fabra, 2010-2012.
- *Foundations of Physics*, Faculty of Computer Science, University Pompeu Fabra, 2006-2011.
- *Experimental Physics*, Faculty of Physics, Imperial College, 2006.
- *Programming in C++*, Faculty of Physics, Imperial College London, 2005-2006.
- *Classical Electrodynamics*, Faculty of Physics, University of Barcelona, 2003-2004.
- *Mechanics Laboratory*, Faculty of Physics, University of Barcelona, 2003-2004.
- *Mathematical Analysis*, Faculty of Physics, University of Barcelona, 2001-2003.

EUROPEAN AND NATIONAL PROJECTS

IP-RACINE (IST-2-511316-IP), Integrated Project Research Area Cinema, 2006-2008. Project devoted to improving the quality of the European digital cinema industry by creating a complete 'from scene to screen' workflow. *Research Lead for Barcelona Media.*

2020 3D Media (FP7-ICT-2007, 2008-2011) is a large IP European project devoted to improving the European industry related to 3D media, including capture, postproduction, distribution, and exhibition. *Research Lead for Barcelona Media.*

i3media (CENIT call, 2007-2010) is a large Spanish research project focused on the creation and automation of intelligent audiovisual content. The audio group investigates and develops new technologies for the creation, postproduction, and exhibition of audio content in an automated way. *Research Lead for Barcelona Media.*

iMP (FP7, 2009-2011) is a STREPs European project aimed at creating architecture, workflow, and applications for intelligent metadata-driven processing and distribution of digital movies and entertainment. *Research Lead for Barcelona Media.*

TCEyF (2000-2004, AEN1998-0431 and FPA2001-3598), aimed at the study of fundamental and effective quantum field theories of gravity and unification of all known basic interactions.

PUBLICATIONS

1. J. Herrera-Joancomarti, C. Perez-Sola, T. Mateos, "Scalable Non-Fungible Tokens on Bitcoin", Cryptology ePrint Archive, [cryptology/2025/641]; 2025.
2. Breebaart, Jeroen; Cengarle, Giulio; Lu, Lie; Mateos, Toni; Purnhagen, Heiko; Tsingos, Nicolas, "Spatial Coding of Complex Object-Based Program Material", Journal of Audio Engineering Society, Volume 67 Issue 7/8 pp. 486-497; 2019.
3. G. Cengarle, T. Mateos, "Effect of microphone number and positioning on the average of frequency responses in cinema calibration", Audio Engineering Society Convention 136, 2014. **Awarded best peer-reviewed paper**.

4. D. Arteaga, D. Garcia, T. Mateos, J. Usher, “Scene Inference from Audio”, Audio Engineering Society Convention 134, 2013.

5. J. Escolano, C. Spa, A. Garriga, T. Mateos, “Removal of afterglow effects in 2-D discrete-time room acoustics simulations”, Applied Acoustics, Volume 74, Issue 6, pp. 818-822, June 2013.

6. G. Cengarle, T. Mateos, “A playback-system-independent clipping detector for multichannel audio production”, AES 132nd Convention, Hungary, April 2012.

7. G. Cengarle, T. Mateos, D. Bonsi, “A second order Ambisonic device using velocity transducers”, JAES Volume 59 Issue 9 pp. 656-668; September 2011.

8. G. Cengarle, T. Mateos, “Comparison of anemometric probe and tetrahedral microphones for sound intensity measurements”, Presented at the 130th AES Convention, London, May 2011.

9. T. Mateos, J. Escolano, C. Spa, A. Garriga, “Compensation of the afterglow phenomenon in 2-D discrete-time simulation”, IEEE Signal Processing Letters, vol. 17, issue 8, p758-761, June 2010.

10. G. Cengarle, T. Mateos, N. Olaiz, P. Arumi, “A new technology for the assisted mixing of sport events: application to live football broadcasting”, Presented at the 128th AES Convention, London, May 2010.

11. D. Garcia, D. Arteaga, J. Usher, T. Mateos, “Determining a room geometry from its impulse response”, Presented at Internoise, Lisbon, June 2010.

12. P. Arumi, N. Olaiz, T. Mateos, “Remastering of movie soundtracks into immersive 3D audio”, Blender Conference 2009, Amsterdam.

13. Toni Mateos, “The arrival of 3D audio”, Keynote in the 6th European Conference on Visual Media Production (CVMP), London, November 2009.

14. G. Cengarle, T. Mateos and D. Bonsi. “Confronto sperimentale tra microfono B-format Soundfield e sonda intensimetrica di pressione e velocit Microflown”, 36th national meeting of the Italian Acoustic Association (AIA), Turin, June 2009.

15. C. Spa, T. Mateos, A. Garriga. “Methodology for studying the numerical speed of sound in finite differences schemes”, Acta Acustica united with Acustica, Volume 95, Number 4, July/August 2009, pp. 690-695.

16. Olaiz, N. Arumi, P. Mateos, T. Garcia, D. 2009. “3D-Audio with CLAM and Blender’s Game Engine” Proceedings of the 7th International Linux Audio Conference (LAC09); April 2009; Italy.

17. Jaume Durany, Adan Garriga, Pau Arumi, Toni Mateos, “Sound spatialization using ray tracing”, Proc. of the EAA Symposium on Auralization, Espoo, Finland, 15-17 June 2009.

18. W. Bailer, P. Arumi, T. Mateos, A. Garriga, J. Durany, and D Garcia, “Estimating 3D Camera Motion for Rendering Audio in Virtual Scenes”, 5th European Conference on Visual Media Production, 2008.

19. Presentation and posters at *ASA Acoustics 2008*, Paris:

    - Pau Arumi, David Garcia, Toni Mateos, Adan Garriga, Jaume Durany, “Real-time 3D audio for digital cinema”, J Acoust Soc Am. 2008 May ;123 (5):3937 18532723.

- Jaume Durany, Adan Garriga, Toni Mateos, “Towards a realistic ray tracing for room acoustics”, J Acoust Soc Am. 2008 May ;123 (5):3769 18532087.
- Carlos Spa, Toni Mateos, Adan Garriga, “General impedance boundary conditions in pseudospectral time-domain methods for room acoustics”, J Acoust Soc Am. 2008 May ;123 (5):3760 18532057.

20. J. P. Gauntlett, O. Mac Conamhna, T. Mateos and D. Waldram, “AdS spacetimes in M-theory,” Fortsch. Phys. **55** (2007) 721.
21. J. P. Gauntlett, O. Mac Conamhna, T. Mateos and D. Waldram, “New supersymmetric AdS(3) solutions,” Phys. Rev. D **74**, 106007 (2006) [arXiv:hep-th/0608055].
22. J. P. Gauntlett, O. Mac Conamhna, T. Mateos and D. Waldram, “Supersymmetric AdS(3) solutions of type IIB supergravity,” Phys. Rev. Lett. **97**, 171601 (2006) [arXiv:hep-th/0606221].
23. J. P. Gauntlett, T. Mateos, O. Mac Conamhna and D. Waldram, “AdS spacetimes from wrapped M5 branes,” JHEP **0611**, 053 (2006) [arXiv:hep-th/0605146].
24. T. Mateos, “Continuous Families of Conformal Field Theories from String Theory,” Fortsch. Phys. **54** (2006) DOI 2005 10288.
25. T. Mateos, “Marginal deformation of N = 4 SYM and Penrose limits with continuous spectrum,” JHEP **0508** (2005) 026 [arXiv:hep-th/0505243].
26. J. P. Gauntlett, S. Lee, T. Mateos and D. Waldram, “Marginal deformations of field theories with AdS(4) duals,” JHEP **0508** (2005) 030 [arXiv:hep-th/0505207].
27. T. Mateos, “D-branes, gauge / string duality and noncommutative theories,” arXiv:hep-th/0409259.
28. T. Mateos, “Supersymmetry of tensionless rotating strings in AdS(5) x S**5,” Fortsch. Phys. **53** (2005) 943.
29. D. Mateos, T. Mateos and P. K. Townsend, “More on supersymmetric tensionless rotating strings in $AdS_5 \times S^5$,” Quantun Theory and Symmetries, Proceedings of the 3rd International Symposium (2003) [arXiv:hep-th/0401058].
30. D. Mateos, T. Mateos and P. K. Townsend, “Supersymmetry of tensionless rotating strings in $AdS_5 \times S^5$, and nearly-BPS operators,” JHEP **0312** (2003) 017 [arXiv:hep-th/0309114].
31. J. Gomis, T. Mateos, P. J. Silva and A. Van Proeyen, “Supertubes in reduced holonomy manifolds,” Class. Quant. Grav. **20** (2003) 3113 [arXiv:hep-th/0304210].
32. J. Brugues, J. Gomis, T. Mateos and T. Ramirez, “Commutative and noncommutative N = 2 SYM in 2+1 from wrapped D6-branes,” Class. Quant. Grav. **20** (2003) S441 [arXiv:hep-th/0212179].
33. T. Mateos, J. M. Pons and P. Talavera, “Supergravity dual of noncommutative N = 1 SYM,” Nucl. Phys. B **651** (2003) 291 [arXiv:hep-th/0209150].
34. J. Brugues, J. Gomis, T. Mateos and T. Ramirez, “Supergravity duals of noncommutative wrapped D6 branes and supersymmetry without supersymmetry,” JHEP **0210** (2002) 016 [arXiv:hep-th/0207091].

35. J. Gomis and T. Mateos, "D6 branes wrapping Kaehler four-cycles," Phys. Lett. B **524** (2002) 170 [arXiv:hep-th/0108080].

36. T. Mateos and A. Moreno, "A note on unitarity of non-relativistic non-commutative theories," Phys. Rev. D **64** (2001) 047703 [arXiv:hep-th/0104167].

37. J. Gomis, K. Kamimura and T. Mateos, "Gauge and BRST generators for space-time non-commutative U(1) theory," JHEP **0103** (2001) 010 [arXiv:hep-th/0009158].

PATENTS

1. "Method and system for intrusion and fire detection", D. Arteaga, U. Sayin, C. Lopez Ullod, T. Mateos, D. Scaini. WO2015074685A1

2. "Upmixing Method and System for Multichannel Audio Reproduction", J. Usher, T. Mateos. EP2614659A1

3. "Method and Apparatus for Three-Dimensional Acoustic Field Encoding and Optimal Reconstruction", T. Mateos, P. Arumi. EP2382803A1; US9299353B2; J20110305344

4. "Method and apparatus for layout and format independent 3D audio reproduction", D. Arteaga, P. Arumi, T. Mateos. WO2013167164A1; J9378747

5. "Method and Device for Automatically Controlling Audio Digital Mixers", F. Jarque Moyano, G. Cengarle, T. Mateos, P. Arumi. EP2421182A1

6. "Automatic loudspeaker polarity detection", M. F. Davis, L. D. Fielder, T. Mateos, G. Cengarle, S. Bharitkar. WO2014116518A1; EP2949133A1; J9560461

7. "Panning of audio objects to arbitrary speaker layouts", T. Mateos, G. Cengarle, D. J. Breebaart, N. R. Tsingos. EP3028476B1; J9712939

8. "Method, Apparatus or Systems for Processing Audio Objects", D. J. Breebaart, L. Lu, N. R. Tsingos, T. Mateos. US20220046378A1; J20250142285

9. "Processing Spatially Diffuse or Large Audio Objects", J. Breebaart, T. Mateos, H. Purnhagen, N. R. Tsingos. JP2024105657A; J10595152

10. "Apparatus and method for reducing temporal artifacts for transient signals in a decorrelator circuit", D. J. Breebaart, L. Lu, T. Mateos, N. R. Tsingos. EP3028274B1; J9747909

11. "Time-Varying Metadata Representation with Lossless Resampling", B. Arnott, J. Breebaart, T. Mateos, D. McGrath, H. Purnhagen, F. Sanchez, N. R. Tsingos. WO2015006112A1; US9858932B2

12. "Geo-Referencing Media Content", T. Mateos. US20150215410A1; J9843642

13. "Methods, apparatus and systems for position-based gain adjustment of object-based audio", N. R. Tsingos, D. McGrath, F. Sanchez, T. Mateos. J11115776

14. "Spatial Error Metrics of Audio Content", J. Breebaart, L. Chen, L. Lu, T. Mateos, N. Tsingos. EP3092642B1; J10492014

15. “Spatial Audio Signal Manipulation”, J. Breebaart, L. Zhu, T. Mateos, H. Purnhagen, N. Tsingos. J12470888; US12470888B2; J20240305945
16. “Methods and Apparatus for Rendering Audio Objects”, T. Mateos, N. R. Tsingos. US20240334145A1; J20240334145
17. “Action Sound Capture Using Subsurface Microphones”. G. Cengarle, T. Mateos, N. Olaiz, K. Hunold. US8335330B2; EP3281416B1; J10136216
18. “Network-based processing and distribution of multimedia content of a live musical performance”, P. Nicol, T. Mateos, G. Cengarle, C. Michel Vasco. US11749243B2; WO2018017878A1; J20220303593
19. “Automatic discovery and localization of speaker locations in surround sound systems”, D. Arteaga, G. Cengarle, D. M. Fischer, T. Mateos, D. Scaini, A. Seefeldt. US11425503B2; J11425503
20. “Hybrid audio signal synchronization based on cross-correlation and attack analysis”, T. Mateos, G. Cengarle. US11609737B2
21. “Audio Input and Output Device with Streaming Capabilities”, G. Cengarle, T. Mateos, D. Scaini, S. S. Barkale. EP3652950A1; J11735194
22. “Automatic gain compensation inference and correction”, T. Mateos, G. Cengarle. WO2019063615A1; J11206001
23. “Audio De-Esser Independent of Absolute Signal Level”, G. Cengarle, T. Mateos, B. G. Crockett. US12051435B2; J12051435
24. “Rendering of Audio Objects with Apparent Size to Arbitrary Loudspeaker Layouts”, T. Mateos, N. R. Tsingos. EP3282716A1; US20180167756A1; J20180167756
25. “Dynamic Equalization”, G. Cengarle, T. Mateos, D. J. Breebaart. US11430463B2; J11430463
26. “Noise Floor Estimation and Noise Reduction”, D. Scaini, T. Mateos, G. Cengarle. US20230081633A1; J12033649
27. “Improved Rendering of Immersive Audio Content”, M. Mason, J. Torres, T. Mateos, D. Arteaga, A. Mills, M. de Burgh, A. Owen. EP4333461A3; US20240305952A1; J20240305952
28. “Rendering audio objects in a reproduction environment that includes surround and/or height speakers”, D. J. Breebaart, T. Mateos, H. Purnhagen, N. R. Tsingos. WO2016040623A1; J20170289724
29. “Method converting multichannel audio content into object-based audio content and a method for processing audio content having a spatial position”, G. Cengarle, T. Mateos. US10863297B2; J10863297
30. “Methods, systems, devices and computer program products for adapting external content to a video stream”, T. Mateos. J11610569; J20210241739
31. “New data, content package, system and method for secure updating in distributed register networks”, T. Mateos, A. Siniscalchi. EP4318266A4; J20240205291
32. “System and method for the secure peer-to-peer transmission of content in distributed ledger networks”, T. Mateos, A. Siniscalchi. MX2022008829A; J20230042916
33. “System and method for dynamic computation scaling in distributed ledger networks”, T. Mateos, A. Siniscalchi. MX2022010110A; J20230085773